SATHISKUMAR SARAVANAN

Full-Stack Software Engineer · MEAN Stack · Immersive Experience

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PROFESSIONAL SUMMARY

Results-driven Full-Stack Software Engineer with 2+ years of experience building scalable digital platforms, immersive AR/VR/WebXR applications, and cloud-based solutions. Skilled in JavaScript, Typescript, Node.js, Python, and modern web technologies, with strong expertise in RESTful APIs, system architecture, real-time applications, and interactive 3D experiences. Experienced in developing production-grade solutions across OTT platforms, AR learning systems, and immersive applications, with a strong focus on performance optimization and user engagement. Published research paper in the area of Augmented Reality with a solid academic background (CGPA 8.46) and a passion for innovation in spatial computing, AI-driven experiences, and next-generation interactive technologies.

EXPERIENCE

Tata Consultancy Services | **Software Engineer (Immersive Experience)** *Feb 2025 – Present · Chennai, India*

- Built a zero-dependency JavaScript tool to auto-generate structured Unity procedure JSON and controller mappings, reducing manual XR content setup time by ~70% and adopted by 15+ internal engineers.
- Engineered a dynamic WebXR AR experience generator using AI agents, MindAR, and A-Frame, enabling automated creation of interactive AR scenes and cutting production time per experience from 3 days to under 4 hours.
- Collaborated with cross-functional teams to define AR content schemas, improving content pipeline consistency and eliminating configuration errors across 20+ deployed experiences.

Shortfundly | **Full-Stack Developer (MEAN Stack)** *Dec 2024 – Feb 2025 · Chennai, India*

- Engineered scalable REST APIs for Shortfundly's OTT platform using Node.js and Express.js, optimizing content delivery throughput and reducing average API response time by ~35% under high-traffic conditions.
- Built a fully responsive Angular frontend with modular component architecture, improving UI load speed and boosting user engagement metrics; navigated 10K+ concurrent user sessions without degradation.
- Architected a multi-tenant white-label OTT platform (Brio) with modular subscription, content management, and branding modules — reducing new client onboarding from 3 weeks to under 5 days across 3+ deployments.

PROJECTS

EduAR — Augmented Reality Learning Platform — *C# · Unity · Vuforia · Android* [GitHub | APK]

- Engineered an interactive AR visualization tool bringing Tamil Nadu state board curriculum (grades 1–10) to life in 3D — enabling students to learn complex spatial and biological concepts through hands-on manipulation rather than static textbook diagrams.
- Integrated Vuforia image tracking with optimized 3D scene hierarchy, achieving <200ms load latency and <100ms frame response for real-time gesture interactions in resource-constrained Android devices.
- Published the research methodology in the Indian Journal of Natural Sciences (2023), contributing to peer-reviewed understanding of AR-enhanced K-12 pedagogy.

WebXR AR Experience Generator — *JavaScript · MindAR · A-Frame · AI Agents*

- Built an AI-agent pipeline that ingests content specs and auto-generates fully functional WebXR AR scenes — accelerating content production 5× compared to manual authoring.
- Designed a schema-driven JSON configuration layer that decouples content from rendering, enabling non-technical authors to create AR experiences without code.

TECHNICAL SKILLS

Languages	C++ · JavaScript / TypeScript · Python · C# · C · HTML · CSS
Backend	Node.js · Express.js · REST APIs · JWT Auth · FastAPI
Frontend	Angular · React · WebXR · A-Frame · MindAR
Databases	MongoDB · MySQL
Tools & Cloud	Git · Docker · AWS · CI/CD · Postman · Linux
AR / XR	Unity · Vuforia · MindAR · A-Frame · WebXR
CS Fundamentals	Data Structures & Algorithms · OOP · Operating Systems · System Design

EDUCATION

University College of Engineering, Ramanathapuram

Graduated June 2024

B.E. in Computer Science · CGPA: 8.46 / 10.0

Relevant coursework: Data Structures & Algorithms · Operating Systems · Object-Oriented Programming · Augmented Reality

ACHIEVEMENTS & AWARDS

- Best Paper Award — NCIDA Conference 2023, for research on AR-based virtual business cards.
- 2nd Place — Tamil Nadu Naan Mudhalvan Niral Thiruvizha 2024 (state-level tech competition).
- Project selected for Final Round — IFERP Project Competition 2023.
- 250+ problems solved on LeetCode with consistent practice in data structures and algorithms.

PUBLICATIONS

S. Sathiskumar, Dr. A. Muthumari, M. Saiban Ali, M. Dinesh. *"A Novel Approach to Virtual Business Cards through Augmented Reality."* Indian Journal of Natural Sciences, 2023.